University of Florida, Department of Mathematics

Algebra First-Year Exam, August 2022, Part 1

Answer 4 questions. Write your solutions clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Prove that the alternating group A_n, $n \geq 3$, is generated by 3-cycles. (You may assume that S_n is generated by transpositions.)

2. This problem has two parts.

(a) Prove that a group G of order 75 has a unique subgroup H of order 25.

(b) Show that if the above subgroup H is cyclic then G is abelian.

3. Let G be symmetric group S_n acting on $X = \{1, 2, \dots, n\}$, $n \geq 2$.

(a) Explain how we have an induced action on the set $X \times X$ of ordered pairs (a, b), with $a, b \in \{1, 2, \dots, n\}$.

(b) Show that there are two orbits of the action on $X \times X$ and describe them.

(c) Describe the stabilizer in G of an element in each of the two orbits.

4. This problem is about $G = \mathrm{GL}(3, 2)$, the group of invertible 3×3 matrices with entries from the field $\mathbb{F}_2$ of two elements.

(a) Compute the order of G.

(b) Prove that the set

$$S = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \middle| \ a, b, c \in \mathbb{F}_2 \right\}$$

forms a Sylow 2-subgroup of G.

(c) Find a subgroup H with $S \lneq H \lneq G$.

5. Let G be a finite group and let p be a prime. A p-subgroup of G is a subgroup whose order is a power of p.

(a) Prove that if P and Q are normal p-subgroups of G, then so is PQ.

(b) Deduce that G has a subgroup K that is a normal p-subgroup and contains every normal p-subgroup.

(c) Prove that G/K has no nontrivial normal p-subgroups.

(d) Give an example in which $|G/K|$ is divisible by p.

6. Let G be a group and α an automorphism of G. Suppose g and h are elements of G with $\alpha(g) = h$. Prove that there exists a group $\widetilde{G}$ such that G is isomorphic to a normal subgroup of $\widetilde{G}$, and the images of g and h in $\widetilde{G}$ are conjugate in $\widetilde{G}$.